

Rank rigidity and the remaining conicity problem for four equal Gaussian cells

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Abstract

For the maximum of the sum of squared Gaussian first moments over four measurable cells of mass $1/4$ in dimension at least three, we prove that every maximizing partition has moment rank exactly three. A second variation in an unused Gaussian coordinate bounds the facet-weight Laplacian by the identity. Its combination with Gaussian flux excludes planar configurations in convex position. For a moment inside the other three moments' convex hull, Gaussian isoperimetry and the elementary three-cell propeller bound give a strict contradiction. We also prove that a maximizing partition which is conical at the origin must be the regular tetrahedral partition, and that the regular partition is locally isolated among all full-rank equal-mass stationary partitions, modulo rotations. These are necessary-condition and conditional-classification theorems, not a resolution of the global equal-mass conjecture. The remaining obstruction is a full-rank, noncentral, irregular tetrahedral partition. No priority or external human-review assertion is made.

1. Setting and exact conclusions

Let γ_d be standard Gaussian measure, with density $\varphi_d(x) = (2\pi)^{-d/2}e^{-|x|^2/2}$. For a measurable partition $\mathcal{C} = (C_1, C_2, C_3, C_4)$ of $\mathbb{R}^d$, put

$$m_i = \int_{C_i} x d\gamma_d(x), \quad F(\mathcal{C}) = \sum_{i=1}^4 |m_i|^2, \quad \gamma_d(C_i) = 1/4. \quad (1)$$

Partitions and boundaries are understood up to Gaussian null sets. The word moment always means the unnormalized vector m_i , not the conditional mean $4m_i$. The moment rank is the dimension of their linear span. Since $\sum_i m_i = 0$, the rank is at most three.

Theorem 1 (rank rigidity). The maximum in (1) is attained. If $d \geq 3$, every maximizing partition has moment rank three. After an orthogonal change of coordinates, its cells are the products with $\mathbb{R}^{d-3}$ of a translated simplicial normal fan in $\mathbb{R}^3$:

$$C_i = \{x : m_i \cdot (x - w) \geq m_j \cdot (x - w) \text{ for all } j\}. \quad (2)$$

Here the four m_i are affinely independent, and w is a unique apex in their span. In particular, the theorem excludes every planar Laguerre diagram, including the case of a bounded central cell.

Theorem 2 (central classification). If a maximizing partition in Theorem 1 is conical at the origin, it is the regular tetrahedral partition, up to rotation and relabeling. Its value is

$$F_{\text{tet}} = \frac{12\theta^2}{\pi^3}, \quad \theta = \arctan \sqrt{2}. \quad (3)$$

In particular, proving Heilman’s 2014 Conjecture 3 (origin conicity of all four-cell equal-mass maximizers) would prove the regular tetrahedral classification. Conversely, that classification implies Conjecture 3. The two assertions are therefore equivalent in this case.

Theorem 3 (local stationary uniqueness). In the space of centered full-rank moment lists and apices, there is a neighborhood of the regular tetrahedral solution in which every equal-mass stationary solution of (2) is a rotation of that solution. A stationary solution here means that the vectors inducing the cells are exactly the cells’ Gaussian first moments. The assertion is local, without an explicit neighborhood radius or a global exclusion.

Moreover the regular root is a strict local maximum of the balanced score objective after removing rotations. Consequently, for all measurable partitions whose normalized moment lists are sufficiently close to a rotated regular list, $F \leq F_{\text{tet}}$, with equality only for the regular partition. This local statement gives no specified size for the neighborhood.

For clarity about sources: Heilman [H14, Conjecture 3, pp.36–37] says *simplicial conical*, not *regular simplicial conical*. His Definition 1.7 distinguishes the two. The arbitrary-mass claim [H19, Conjecture 1.16] has a counterexample in the companion note [EX]; it does not settle (1). The unit-variance Gaussian maximum comparison [M26] has no general cell prices and is not used in these proofs.

2. Attainment, prices, and Gaussian flux

We include the structural argument to specify the class covered by the subsequent variation. Let $p_i > 0$ sum to one. Relax a partition to functions $0 \leq f_i \leq 1$, $\sum_i f_i = 1$, $\int f_i d\gamma = p_i$. This is a weakly compact convex subset of $L^2(\gamma)^4$. Its moment maps are weakly continuous because every coordinate of x is in $L^2(\gamma)$. The moment objective therefore attains its maximum on the relaxed set. If m_i are the moments of a relaxed maximizer, every feasible f has $\sum_i m_i \cdot \int x f_i \leq \sum_i |m_i|^2$: otherwise expanding the squared moments gives a strictly larger value. Consequently the maximizers of this linear functional form an exposed face. Every member of the face also maximizes the squared objective, by Cauchy–Schwarz and the definition of the maximum. The face has an extreme point by the Krein–Milman theorem, which is also extreme in the relaxed set. Its functions are indicators: if two functions are bounded away from both zero and one on a set of positive measure, nonatomicity gives two equal-measure subsets there; adding a small opposite signed transfer on these subsets produces two distinct feasible functions whose average is the original. There are finitely many pairs, so any nonindicator tuple admits such a transfer. This proves attainment by a measurable partition.

At any maximizing partition the m_i are distinct. Indeed equal moments in two positive-measure cells allow an equal-mass exchange of small sets near distinct density points. Their integral difference $\Delta \neq 0$ increases the objective by $2|\Delta|^2$.

For distinct scores v_i , define

$$\Psi(v, \lambda) = \sum_i p_i \lambda_i + \mathbb{E} \max_i (v_i \cdot G - \lambda_i), \quad C(v) = \min_{\lambda} \Psi(v, \lambda). \quad (4)$$

Fix $\min_i \lambda_i = 0$. Then $\Psi \geq p_{\min} \max_i \lambda_i$, so a minimum exists. Affine-score ties have Gaussian measure zero, and differentiating under the integral gives $\partial_{\lambda_i} \Psi = p_i - \gamma(C_i)$. Thus its minimizing prices give precisely the prescribed masses. The corresponding Laguerre cells maximize the linear assignment $\sum_i v_i \cdot \int_{C_i} x$. Every global squared-objective maximizer also maximizes this linear assignment with $v_i = m_i$, by the expansion just used. Equality in the pointwise maximum then identifies its cells as these Laguerre cells. All inequalities depend only on the span of the moments; the other Gaussian coordinates factor out. This proves the cylindrical assertion before the rank is identified.

Write Σ_{ij} for the relatively open ij facet, excluding multiple ties, and set zero for an absent facet. In the effective space of rank r , define

$$A_{ij} = \int_{\Sigma_{ij}} \varphi_r d\mathcal{H}^{r-1}, \quad w_{ij} = \frac{A_{ij}}{|m_i - m_j|}, \quad (Lu)_i = \sum_{j \neq i} w_{ij}(u_i - u_j). \quad (5)$$

The symbol w_{ij} denotes an edge weight; the bold geometric apex in (2) is the vector w . Formula (5) applies also to cylindrical facets because the extra Gaussian coordinates integrate to one. Gaussian integration by parts gives

$$m_i = \sum_{j \neq i} w_{ij}(m_i - m_j), \quad LM = M, \quad (6)$$

where M is the matrix whose rows are m_i . For justification on unbounded cells, intersect with balls of radius R : the artificial Gaussian boundary flux is $O(R^{r-1}e^{-R^2/2})$, tending to zero. The inward ij normal is $(m_i - m_j)/|m_i - m_j|$.

3. The unused-coordinate second variation

Lemma 4 (spectral constraint). If a maximizing partition has an unused Gaussian coordinate, its matrix (5) satisfies

$$L \preceq I \text{ on } \mathbf{1}^\perp, \quad W := P - L \succeq 0, \quad P = I - \mathbf{1}\mathbf{1}^T/4. \quad (7)$$

Proof. Let X be the effective Gaussian vector and Z an independent standard normal in an unused direction e . Perturb scores to $v_i(t) = m_i + ta_i e$. The balanced linear assignment value is $C(v(t))$. For every v , completing squares cell by cell proves

$$H(v) := 2C(v) - \sum_i |v_i|^2 \leq \max F, \quad H(m) = F. \quad (8)$$

At fixed prices, the first variation in t vanishes by $\mathbb{E}Z = 0$. The second variation of the expected maximum is

$$\partial_{tt} \Psi(v(t), \lambda)|_{t=0} = \sum_{i < j} \frac{1}{|m_i - m_j|} \int_{\Sigma_{ij}} (a_i - a_j)^2 \mathbb{E}Z^2 \varphi_r d\mathcal{H}^{r-1} = a^T L a. \quad (9)$$

There is no extra factor two in (9). Locally across one facet, if s is its signed normal distance, the difference of the two scores is $|m_i - m_j|s$. Integrating the change in the maximum after shifting this

difference by $t(a_i - a_j)Z$ gives $t^2(a_i - a_j)^2 Z^2 / (2|m_i - m_j|)$ times the surface density to second order. The second derivative is (9).

We spell out why optimizing the prices does not change (9). The price Hessian at zero is L , obtained by the same facet calculation. Its kernel is precisely the common-price direction. The facet adjacency graph is connected: a generic segment between interiors of any two positive cells crosses only ordinary facets. Hence L is positive definite on $\mathbf{1}^\perp$. The mixed derivatives $\partial_t \partial_\lambda \Psi$ vanish because the first-order flux is proportional to $\mathbb{E}Z$. The implicit function theorem for the mass equations therefore gives $\lambda'(0) = 0$ on the common-price gauge slice. Along these minimizing prices, (9) remains the second derivative of C .

The differentiability used here holds also when four planar cells meet at one point. Indeed no triple tie can have codimension one when all cells have positive mass: in that event three slopes are affinely collinear with matching prices, and the middle score is a convex combination of the other two everywhere, so its cell has zero mass. All multiple ties thus have codimension at least two. On compact sets, a tube of thickness $O(t)$ around these ties has volume $O(t^2)$, while a score perturbation is $O(t)$, so it changes the expected maximum only by $O(t^3)$. Away from the tubes the two-score facet calculation applies. Gaussian tails make the same estimates integrable for the linear factors $1 + |X| + |Z|$; one may first restrict these factors, apply the compact calculation, then let the restriction tend to infinity. This proves the C^2 expected maximum formula and the C^1 mass map used in the implicit function argument.

At the maximum, (8) has nonpositive second derivative. Its value is $2(a^T L a - |a|^2)$, so $L \preceq I$. Since $L\mathbf{1} = 0$, this is equivalent to $P - L \succeq 0$. $\square$

4. A self-contained three-cell bound and a strict profile estimate

Put $B = 9/(8\pi)$. We use the classical three-cell bound

$$\sum_{i=1}^3 \left| \int_{D_i} x d\gamma_d \right|^2 \leq B \quad (10)$$

without mass restrictions. Here is a short proof. If its moments b_i have squared sum $S > 0$, their centered scores $v_i = b_i/\sqrt{S}$ span at most two dimensions and have squared norm sum one. Pointwise maximization gives $\sqrt{S} \leq \mathbb{E} \max_i v_i \cdot G$. The maximizing score cells are at most three planar sectors, possibly with empty cells. For a sector of angle α , its moment is $\sin(\alpha/2)/\sqrt{2\pi}$ times its bisector. Thus the sector partition has squared moment sum $\sum_i \sin^2(\alpha_i/2)/(2\pi)$. If $x_i = \alpha_i/2$ sum to π , the identity $\sum \sin^2 x_i = 2 + 2 \prod \cos x_i$ bounds this by $9/(8\pi)$: a negative cosine makes the product nonpositive, and otherwise concavity of $\log \cos$ bounds the product by $1/8$ (endpoints by continuity). Cauchy–Schwarz on the score-cell moments now gives $\mathbb{E} \max_i v_i \cdot G \leq \sqrt{B}$, proving (10). Rank-one scores are handled by the same sectors after adjoining an unused coordinate. This proof is a restatement of the classical propeller calculation [KN09], not a novelty assertion.

Let $q = \Phi^{-1}(3/4)$ and $h_0 = \phi(q)$, where $\phi(t) = (2\pi)^{-1/2} e^{-t^2/2}$ and Φ is its distribution function. We need the strict elementary estimate

$$h_0^2 > \frac{9}{32\pi} = \frac{B}{4}. \quad (11)$$

Indeed $\pi < 22/7 < 32/9$ gives $\phi(0) > 3/8$, and $e^{-x^2/2} \geq 1 - x^2/2$ yields $\Phi(3/4) > 1/2 + (3/8)(87/128) > 3/4$. Thus $0 < q < 3/4$. Taylor's theorem gives $e^{-u} \geq 1 - u + u^2/2 - u^3/6$ for $u \geq 0$. This last polynomial is strictly decreasing, so $e^{-q^2} > 4637/8192 > 9/16$, proving (11).

The other input is the classical Gaussian isoperimetric theorem: a polyhedral cell of mass p has total Gaussian perimeter at least $\phi(\Phi^{-1}(p))$. We use it only for a single cell of mass $1/4$. This standard theorem is an imported analytic dependency, not formalized or reproved here; see [B75, ST78] and [H19, §1].

5. Complete exclusion of moment rank two

Suppose a maximizer has rank two. Since $d \geq 3$, Lemma 4 applies. The kernel of W contains the three-dimensional span of $\mathbf{1}$ and the two columns of M . Therefore

$$W = caa^T, \quad c \geq 0, \quad a \perp \mathbf{1}, \quad a^T M = 0. \quad (12)$$

The nonzero vector a spans this one-dimensional orthogonal complement. Comparing off-diagonal entries gives

$$w_{ij} = 1/4 + ca_i a_j. \quad (13)$$

Case A: all four moments are vertices of their convex hull. They form a convex quadrilateral. Its unique affine dependence has, after labeling, $a_1, a_3 > 0$ and $a_2, a_4 < 0$. Write $s_i(x) = m_i \cdot x - \lambda_i$. The number $K = \sum_i a_i s_i(x) = -\sum_i a_i \lambda_i$ is independent of x . On an ordinary 13 facet, $s_1 = s_3 = T > s_2, s_4$ implies $K = a_2(s_2 - T) + a_4(s_4 - T) > 0$. On a 24 facet the same argument gives $K < 0$. Both diagonal facets consequently cannot be present. But (13) gives $w_{13}, w_{24} \geq 1/4$, a contradiction.

Case B: a moment lies in the convex hull of the other three. Relabel it as $m_4 = \sum_{j=1}^3 \alpha_j m_j$, with $\alpha_j \geq 0$ and $\sum \alpha_j = 1$. In (12) use $a = (\alpha_1, \alpha_2, \alpha_3, -1)$. Formula (13) gives $w_{4j} \leq 1/4$, also when an α_j vanishes. Put $t = |m_4|$ and $F = \sum |m_i|^2$. The perimeter of the fourth cell satisfies

$$h_0 \leq \sum_j A_{4j} \leq \frac{1}{4} \sum_{j=1}^3 |m_4 - m_j| \leq \frac{\sqrt{3}}{4} \sqrt{F + 4t^2}. \quad (14)$$

The last identity before Cauchy–Schwarz is $\sum_j |m_4 - m_j|^2 = F + 4t^2$, using $\sum_i m_i = 0$.

If $t > 0$, some j has $m_4 \cdot m_j \geq t^2$ by the convex combination. Merging those two cells and using (10) gives $F + 2t^2 \leq B$. Project onto m_4/t : one projection equals t , another $s \geq t$, and the other two sum to $-t - s$. Hence $F \geq t^2 + s^2 + (t + s)^2/2 \geq 4t^2$. If $t = 0$, the same two inequalities follow by merging any cell with cell four. In either case $t^2 \leq B/6$ and therefore

$$F + 4t^2 \leq B + 2t^2 \leq 4B/3. \quad (15)$$

Equations (14)–(15) imply $h_0 \leq \sqrt{B}/2$, contradicting (11). This includes a moment on an edge of the hull, not only a strictly interior moment. Cases A and B exhaust four distinct rank-two points. Thus rank two is impossible.

6. Rank one, and completion of Theorem 1

If the moment rank is one, distinct slopes give four ordered intervals in that coordinate. Their masses force endpoints $-q, 0, q$. Their moments, in increasing order, are $-\phi(q), \phi(q) - \phi(0), \phi(0) - \phi(q), \phi(q)$. Since $0 < \phi(q) < \phi(0)$, their squared sum is strictly below $2\phi(0)^2 = 1/\pi$. The partition into four coordinate quadrants in a plane has equal masses and value $1/\pi$, so the rank-one partition cannot maximize. Rank zero contradicts distinct moments. This completes the rank conclusion. Rank three and $\sum_i m_i = 0$ make the four points affinely independent. The differences of three prices consequently determine the unique apex w in (2).

An additional useful consequence of (6), requiring no unused coordinate, is

$$L = P, \quad w_{ij} = 1/4 \quad (i \neq j) \quad (16)$$

at every full-rank stationary solution. Indeed the columns of M span $\mathbf{1}^\perp$, on which L is the identity by (6). Conversely (16) and the flux formula make the inducing vectors exactly the cell moments. Thus the six weight equations are exact stationarity equations, not merely necessary scalar inequalities.

7. Central cones: disphenoid geometry and regularity

If a full-rank maximizing partition is conical at the origin, each of its affine facets must contain the origin. All six facets of a full-rank four-score normal fan are present; hence all prices are equal and its apex is zero.

The normal cone at each vertex of the score tetrahedron $T = \text{conv}(m_1, m_2, m_3, m_4)$ has spherical area π , because its Gaussian mass is $1/4$. By the spherical triangle area formula, this normal-cone area is 2π minus the sum of the three planar face angles at that vertex. Thus the sum of face angles at every vertex of T is π .

Lemma 5 (elementary unfolding). A nondegenerate tetrahedron with face-angle sum π at every vertex has equal opposite edges.

Proof. Choose face ABC , and unfold the three adjacent faces containing the remaining vertex D outward into the plane of ABC . Let P, Q, R be the respective images of D from faces ABD, ACD, BCD . At A the consecutive angles from AP to AB , from AB to AC , and from AC to AQ sum to π . Thus P, A, Q are collinear with A between P, Q ; the equal lengths $AP = AQ = AD$ make A their midpoint. Likewise B is the midpoint of PR , and C of QR . The midpoint theorem therefore gives $AD = BC$, $BD = AC$, and $CD = AB$. $\square$

Such a tetrahedron is a disphenoid. Its centered vertices can be written, in orthogonal coordinates,

$$(a, b, c), (a, -b, -c), (-a, b, -c), (-a, -b, c), \quad a, b, c > 0. \quad (17)$$

For completeness, the three vectors $(m_1 + m_2)/2, (m_1 + m_3)/2, (m_1 + m_4)/2$ give this representation; equality of opposite edges makes them pairwise orthogonal. Rank three makes each nonzero. Their signs can be absorbed into the axes.

On the 12 facet, set $s_x = \sqrt{b^2 + c^2}$ and use orthonormal coordinates $X = x, Y = (cy - bz)/s_x$. The remaining inequalities are $X \geq bc|Y|/(as_x)$. Consequently

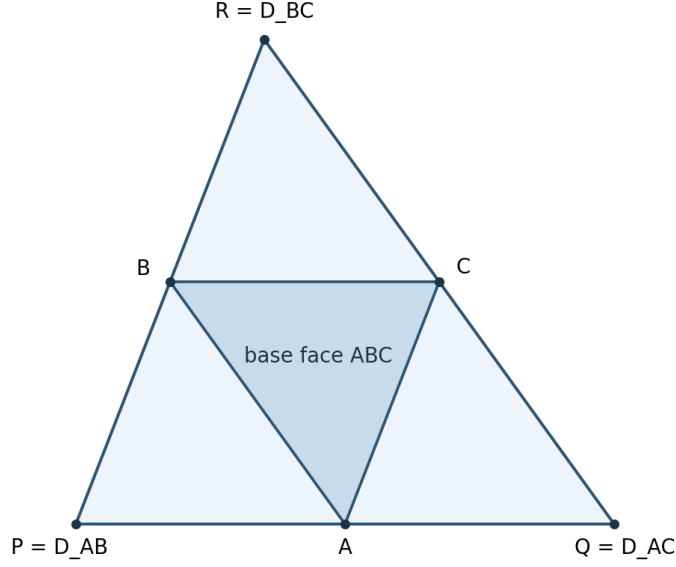


Figure 1: Schematic unfolding: the base face is the medial triangle ABC; P, Q, R are the three images of D.

$$A_{12} = \frac{\phi(0)}{\pi} \arctan \frac{as_x}{bc}, \quad w_{12} = \frac{\phi(0)}{2\pi} \frac{\arctan(as_x/(bc))}{s_x}. \quad (18)$$

The 13 formula exchanges a, b , with $s_y = \sqrt{a^2 + c^2}$. If $a > b$, then $s_x < s_y$ and $as_x/(bc) > bs_y/(ac)$, since the difference of the squared cross-multiplied quantities is

$$(a^2 s_x)^2 - (b^2 s_y)^2 = (a^2 - b^2) \{a^2 b^2 + c^2(a^2 + b^2)\} > 0. \quad (19)$$

Strict monotonicity and positivity of arctangent give $w_{12} > w_{13}$, contradicting (16). Interchanging axes proves $a = b = c$. Formula (18) and (16) now force $a = \theta/\pi^{3/2}$ and $F = 12a^2$, proving Theorem 2. No theorem comparing general Gaussian maxima is needed.

8. Local isolation in all nine parameters

Use the regular solution $m_i^0 = a\sigma_i$, where the four σ_i are the sign vectors in (17), $a = \theta/\pi^{3/2}$. Every centered full-rank list sufficiently close to it is obtained by a linear map. Polar decomposition removes rotations, leaving a positive symmetric map $I + B$, where

$$B = \begin{pmatrix} u & b_z & b_y \\ b_z & v & b_x \\ b_y & b_x & s \end{pmatrix}.$$

The apex perturbation is $z = (z_x, z_y, z_z)$. Thus there are nine parameters. The equations are the three independent equal-mass equations and the six equations $w_{ij} = 1/4$. Their coefficients are

smooth near the regular solution because all cell inequality matrices are nonsingular there. Define

$$c_0 = \frac{\sqrt{2}}{3\theta}, \quad h = \frac{\sqrt{\pi/3}}{\theta}. \quad (20)$$

Here h is the conditional X mean of the planar Gaussian wedge $X \geq |Y|/\sqrt{2}$. Its mass is θ/π and its unnormalized X moment is $\phi(0)\sqrt{2/3}$. The first mass derivatives are

$$\delta p_i = a \sigma_i \cdot (h(b_x, b_y, b_z) - z). \quad (21)$$

To verify (21), on the 12 facet its conditional point mean is $he_x = (h/(2a))(m_1^0 + m_2^0)$. Summing normal fluxes with weights 1/4 gives $\delta p_i = (h/(2a))((m_i^0)^T B m_i^0 - a^2 \text{tr } B) - m_i^0 \cdot z$. Substituting the four sign vectors gives (21).

For the two opposite facets in the x pair the first weight derivatives are

$$\begin{aligned} 4\delta w_{12} &= K_x + (c_0 - 1)b_x - hz_x, \\ 4\delta w_{34} &= K_x - (c_0 - 1)b_x + hz_x, \\ K_x &= c_0\{u - (v + s)/2\} - (v + s)/2. \end{aligned} \quad (22)$$

The other pairs are obtained by cycling coordinates. Here is an explicit check of the geometric derivatives in (22). A diagonal perturbation changes the wedge angle in (18) by $(\sqrt{2}/3)\{u - (v + s)/2\}$ and its edge length by a relative $(v + s)/2$. For a b_x perturbation, the 12 facet stays $y + z = 0$ and its wedge is $X \geq (1 - b_x)|Y|/\sqrt{2}$ to first order, giving angle derivative $(\sqrt{2}/3)b_x$ and relative edge derivative b_x . The opposite facet has opposite signs. The other two off-diagonal perturbations have zero first angular derivative on this pair by its reflection symmetries. Translating the apex changes the surface Gaussian integral by $-A_{12}hz_x$, and by its negative on the opposite facet, because the Gaussian density derivative is $-x \cdot z$. Together these give exactly (22).

If the linearized equations vanish, (21) gives $z = h(b_x, b_y, b_z)$. The opposite-facet sums give $K_x = K_y = K_z = 0$. Their sum is $-(u + v + s)$; with trace zero, $K_x = (3c_0 + 1)u/2$, and similarly for v, s . Thus all three diagonal variables vanish. The differences in (22) give $(c_0 - 1 - h^2)b_x = 0$ and its two cyclic counterparts. But $c_0 < 1$, since $\arctan \sqrt{2} > \sqrt{2}/(1 + 2)$ by its defining integral. Their common coefficient is strictly negative, so the remaining variables vanish as well. The nine-equation derivative is invertible. The implicit function theorem now proves Theorem 3. This proof gives local stationary uniqueness, not a global stability estimate or a computable exclusion radius.

For completeness we also verify the asserted local maximum, rather than deducing it merely from stationary uniqueness. Let $b(M)$ be the actual moments of the balanced score partition generated by M , and $H(M) = 2C(M) - |M|_F^2$ as in (8). The envelope derivative is $\nabla H = 2(b(M) - M)$. Gaussian flux gives $b(M) = L(M)M$. At the regular root $L = P$, so $\delta b - \delta M = \delta L M^0$. Therefore $\delta^2 H = 2 \sum_{i < j} \delta w_{ij}(\delta m_i - \delta m_j) \cdot (m_i^0 - m_j^0)$. Write $u = \rho + u_0$, $v = \rho + v_0$, $s = \rho + s_0$, with $u_0 + v_0 + s_0 = 0$. Substitute (22) and the mass derivative $z = h(b_x, b_y, b_z)$ to obtain

$$\delta^2 H = -24a^2\rho^2 - 2a^2(3c_0 + 1)(u_0^2 + v_0^2 + s_0^2) - 8a^2(1 + h^2 - c_0)(b_x^2 + b_y^2 + b_z^2). \quad (24)$$

All three coefficients are strictly negative. Smoothness and Taylor's theorem show that H is strictly smaller than its regular value F_{tet} for every nearby nonregular score list, modulo rotations. This does not assume a stationary competitor.

Now take any measurable partition with moment list b and $F(b) > 0$, and set $v = \sqrt{F_{\text{tet}}} b / \sqrt{F(b)}$. If this normalized list is in the preceding neighborhood, then $H(v) \leq F_{\text{tet}}$ and $|v|_F^2 = F_{\text{tet}}$ imply $C(v) \leq F_{\text{tet}}$. The original partition is feasible for the linear assignment, so $\sqrt{F_{\text{tet}} F(b)} \leq C(v)$ and hence $F(b) \leq F_{\text{tet}}$. Equality forces v to be a rotated regular list and the partition to attain its linear assignment. Its distinct scores have null ties, forcing the regular cells themselves. This proves the local assertion for all measurable competitors, not only Laguerre diagrams.

9. The precise unclosed global step

There is an explicit compact search domain in the score-and-price coordinates, even though a geometric apex could diverge when a score tetrahedron flattens. At a global maximizer,

$$|m_i| \leq h_0, \quad |m_i - m_j| \geq \frac{1}{16\phi(0)}, \quad \min_i \lambda_i = 0, \quad \max_i \lambda_i \leq 2qh_0. \quad (25)$$

Here is a direct proof, included to make a prospective finite verification concrete. The moment bound follows by comparing the linear integral in direction $m_i/|m_i|$ with the upper halfspace of mass $1/4$, whose moment is h_0 . For a pair of cells, their union D has mass $M = 1/2$. Project D onto any unit coordinate; its Gaussian subdensity g is at most $K = \phi(0)$. Splitting at its median gives two cells of mass $1/4$, with projected moment difference $\int |x - b|g(x) dx$. The layer-cake formula bounds this below by $\int_0^{M/(2K)} (M - 2Ks) ds = M^2/(4K) = 1/(16\phi(0))$. The original pair must have moment difference at least as long: its total moment is fixed and its squared objective is half the sum of the squared total and squared difference. Otherwise this pair replacement improves the full partition. This proves the separation bound. Finally, each cell is contained in its pairwise winning halfspace. Its mass $1/4$ and that of the other cell imply $|\lambda_i - \lambda_j|/|m_i - m_j| \leq q$. The moment bound gives the last assertion of (24).

These bounds protect pair-score denominators and give bounded prices without assuming a bounded apex. They do not bound the smallest singular value, and do not provide a coverage certificate or a global objective bound.

The preceding theorems reduce the original problem to excluding full-rank solutions (2) with nonzero apex. Such a solution has four equal masses and all six ratios $A_{ij}/|m_i - m_j| = 1/4$. It cannot approach the regular solution except by rotation. A sufficient stronger statement would be that these nine equations have no noncentral solution anywhere. We have not proved that statement, nor ruled out noncentral stationary saddles or noncentral global maxima. Proving only the local theorem or finding many numerical roots at the regular solution does not complete this step.

The accompanying discovery code evaluates the equations by one-dimensional Plackett quadrature. For a positive definite three-dimensional cell correlation matrix R and standardized threshold vector t , interpolate $R(s) = I + s(R - I)$. Then its Gaussian CDF is the independent product plus the integral of

$$\sum_{i < j} R_{ij} \phi_2(t_i, t_j; sR_{ij}) \Phi((t_k - \mu_{k|ij}(s))/\tau_{k|ij}(s)), \quad (23)$$

where k is the remaining index and the conditional mean and variance are those of the Gaussian with covariance $R(s)$. Facet integrals are the normal density times an analogous bivariate conditional CDF. The code fixes two batches of 24 and 32 starts; every recorded solve converges numerically to the regular root. One broad run produces an integration warning. No quadrature bounds, interval certificate, coverage argument, or proof of global uniqueness is claimed from these runs.

Lean checks the finite algebraic obstructions and the invertible linearized-system argument. The Gaussian variation, isoperimetry, unfolding geometry, analytic derivatives and global statements are written mathematics, not a whole-paper Lean formalization.

References and attribution

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Finite comparison with these sources did not locate the fixed-mass rank-three theorem or the present combination of classification and local isolation. This is not an exhaustive priority determination.